

Wearable Alert Device: Hardware Design Document

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Senior Capstone Project

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Table of Contents

Overview	2
Power Supply	3
Micro-Lipo Charger	4
Position Module	5
Analog Microphone	6
Arduino Nano Esp32	7
Testing LED	8
Boost Converter	9
Hardware Documentation	11
Block Diagram	11
Schematics	13

Overview

This hardware design document describes the various hardware components used in the building and testing of the CCEW WAD capstone project. This document will list all hardware developments including listings on all parts/chips used in the design of the WAD, Block diagrams of the WAD, Schematics of the WAD, and the final PCB design that was not implemented but planned for.

The WAD's mission is to create a discrete device that could be worn in case of emergency assistance. The overall goal of the WAD is to alert authorities on location, elevation, and an immediate threat such as a gunshot.

The intended audience that we are building the device for is not that of first responders. Instead we are thinking about secondary responders who may not have field training but may still be in dangerous scenarios while working. This device is made for them to have a way of contacting authorities while not giving too much information to the potential assailant.

Power Supply



Description:

A lithium-ion battery pack made of two 2200mAh cells totaling to 4400mAh operating at 3.7V

Specifications:

Parameters:	Values
Lithium-ion Cells	2 x 2200mAh
Max Recommended Charging Rate	1A
Max Recommended Discharging Rate	2.2A

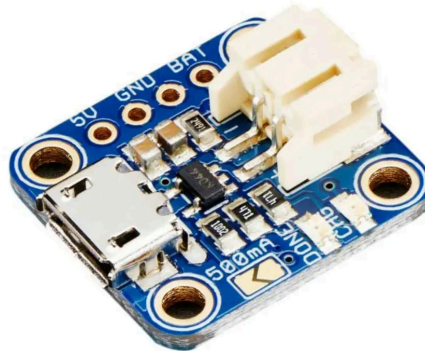
Role in this system:

This battery pack is the main source of power for the device. Connected with a JST 2-pin cable, the pack is compatible with any JST or similar connector.

Why was this component chosen:

This battery pack was chosen as it provides more than enough amperage to cover all components operating within the device. Its milliamperere-hours can power the device for an extended amount of time before needing to recharge. This battery pack is also compatible with the device's chosen charging component while non-Adafruit batteries are not.

Micro-Lipo Charger



Description:

A small Lipo charger able to recharge any 3.7/4.2V lithium polymer or lithium ion rechargeable battery by a JST plug through a Micro-B USB. A red LED will be active when charging and a green LED will be active once charging is complete.

Specifications:

Parameters:	Values
Recommended Batteries	Rechargeable 3.7/4.2V LiPoly/LiIon Batteries (Adafruit Only)
Micro-B USB Connector Voltage	5V
Charge Current	100mA, Adjustable to 500mA

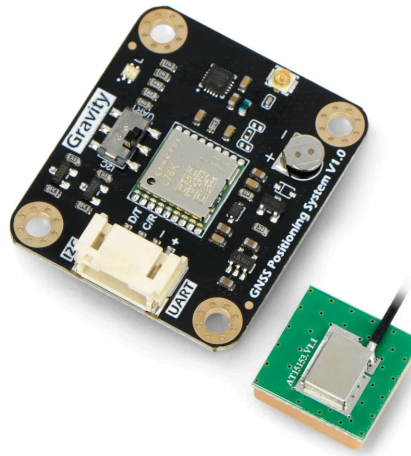
Role in this system:

The Lipo-Charger regulates the charging process using a constant current, prevents overcharging, and protects the battery. Due to the dual functionality of the USB and JST ports, it allows the device to be recharged without disconnecting the battery, offering continuous operation and usability.

Why this component was chosen:

The Micro-Lipo Charger was chosen mainly for its compatibility with the battery pack. It allows the user to be able to charge the device without needing to disconnect the battery. The charger also handles all battery protection independently of the microcontroller without external components, improving safety and reliability. Additionally, the charge current can be changed between 100mA and 500mA or be fully customizable with a single resistor on the PROG pin

Position Module



Description:

The Gravity: Receiver Module is a joint positioning multi-satellite device that provides high-precision, high-speed, and stable data readings of longitude, latitude, altitude, and time.

Specifications:

Parameters:	Values
Operating Voltage	3.3-5.5V DC
Output Signal	I2C/UART
Power Consumption	40mA
Positioning Time	30s-Cold Start; 2s-Hot Start
Antenna Interface	IPEX1
Antenna Frequency	1561-1575.42MHz +/- 3MHz

Role in this system:

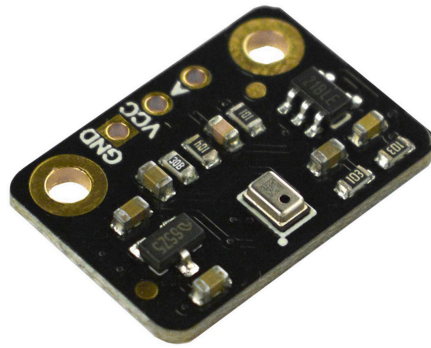
The receiver module is responsible for determining the device's real-time geographical location. It receives signals from multiple satellite constellations and computes position data. This information is sent to the microcontroller where it's constantly updated until an alert is activated and sends out the most recent data to the app for emergency services.

Why this component was chosen:

This module was chosen because it supports multiple satellite constellations, which improves accuracy and reliability in multiple environments. The module offers two-meter accuracy and quick positioning updates, which is critical for emergency response applications. Altitude is given alongside time and coordinates, which some smartphone maps do not offer or

have the information split between different apps. Additionally, the module supports I2C, allowing for simple integration for the microcontroller and reduces wire complexity.

Analog Microphone



Description:

The Fermion MEMS Microphone is an compact analog microphone built on a Micro-Electro-Mechanical System (MEMS).

Specifications:

Parameters:	Values
Operating Voltage	3.3V DC
Direction	Omnidirectional
Output Type	Analog
Maximum Sound Pressure Level	110 dB
Signal to Noise Ratio	59 dB

Role in this system:

This microphone is the main source where soundwaves are read and analyzed. It constantly monitors the environment and produces an analog voltage that will be sent via the amplitude of incoming sounds. This signal will be converted and passed through an ADC GPIO pin on the Arduino Nano ESP32, where it is sampled in the software. This microphone does not perform any type of action or decision-making on its own. The microphone's purpose is to read in the data for the software to read and interpret. In this case, it will read in any type of noise, ranging from "quiet" to a gunshot.

Why this component chosen:

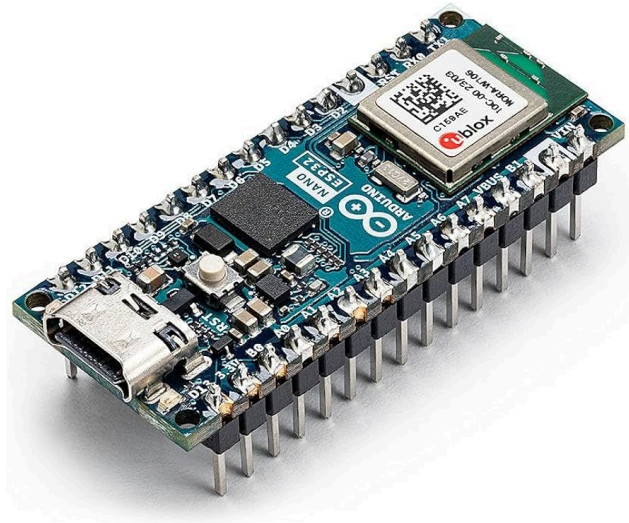
The Fermion MEMS Microphone was chosen based on a misordering issue. Originally, the plan was to use the Fermion I2S MEMS Microphone, which is a digital microphone able to read up to 140 dB. However, there was an ordering issue, and the wrong component came in. The digital microphone would have been using I2S and would have reduced a lot of unnecessary noise; however, the Fermion MEMS Microphone still functions as needed. The MEMS Microphone is directly connected to the ESP32's ADC and has no protocol, meaning no clock or synchronization is needed, making the firmware simpler. The max 110 dB also allows the device to read in all types of loud sounds without damaging the microphone sensors, even if clipping were to occur.

Wiring to the Arduino Nano ESP32:

VCC	3.3V	
GND	GND	Shared GND for Device
Aout	DC GPIO Pin	Analog Audio

A 0.1 uF decoupling capacitor should be placed between the VCC and GND pins of the microphone. This filters out the noise the analog microphone produces before audio sampling.

Arduino Nano Esp32



Description:

The Arduino Nano ESP32 is built around the r-block NORA-W106 module that has capabilities of Wi-Fi and Bluetooth 5.0 functions in a small form factor. This microprocessor will be considered as the brain that will collect the audio from the microphone, run through the firmware, collect GPS data, and transmit an emergency response via Bluetooth. This Arduino

Nano ESP32 can be powered by a USB-C built on the board but this product will be powered by a 3.7V Li-ion battery, running on an operating voltage of 3.3V.

Specifications:

Parameters:	Values
Operating Voltage	3.3V
Bluetooth	Bluetooth 5.0 (Blue Energy)
GPIO	Multiple digital and analog I/O pins
Form	Arduino Nano

Role in the System:

The Arduino Nano ESP32 will serve as the brain of the gunshot-detection and GPS module. When working with the microphone, it will continuously sample voltage data from the microphone output at a defined sampling rate. Each sample will be stored for processing. There will be a Hamming window and the Fast Fourier Transform that will be during this analysis. When the characteristics are met (the threshold), the firmware will send a signal in the event of a gunshot and trigger an alert response. For the button, it will be connected to a GPIO pin to act as an emergency button, regardless of the microphone detection. Upon detection or a button press, the ESP32 will send a signal via Bluetooth to alert responders. The LED connected to the GPIO pin is for testing purposes only, and whenever a detection happens or a button is pressed, it will send a visual instead of looking at the phone.

Why was this component chosen:

The Arduino Nano ESP32 was chosen because it had all the features needed for our product to function. It was sufficient enough to perform Bluetooth communication, processing, and hardware integration.

Testing LED

Description:

The testing LED is a standard red LED used for development and validation assistance. The LED is also used to turn on if there is a gunshot detected or if the button is pressed to signal to the user discreetly that a signal has been sent. This LED will be incorporated into the final product, but it will not be a part of the interface for the user to use. For this product, it is made to be as discreet as possible, so having an LED turn on in the field would potentially alert the perpetrators.

Specifications:

Parameters:	Values
Type	Red LED
Forward Voltage	~2.0V – 3.3V LED (2V for Red)
Current Limiting Resistor	1k Ohm
Control Type	Digital GPIO Pin

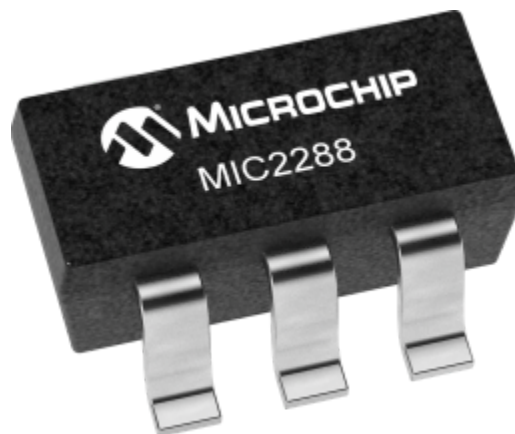
Role in the System:

The LED is connected to the GPIO output on the Arduino Nano ESP32. In the firmware, whenever a gunshot is detected or if the button is pressed, it will send an alert via Bluetooth to the phone to send a signal. During the testing phase, like loud clapping and gunshot, the LED turns on. This allows us to know whether a signal was sent or not and to validate the performance without waiting for the notification on a phone.

Connections:

LED Anode (+)	To a resistor, current limiting, to GPIO Pin
LED Cathode (-)	Directly to GND

Boost Converter



Description:

The MIC2288 is a 1.2MHz PWM, DC/DC boost switching regulator. It implements a constant frequency current-control scheme designed for high-efficiency power conversion.

Specifications:

Parameters:	Values
Input Voltage	2.5-10V
Output Voltage	Adjustable to 34V
PWM Operation	1.2MHz
Switch Current	Over 1A

Role in the system:

The MIC2288 Boost Converter steps up the battery voltage from 3.7V to 5V required by the microcontroller. It provides a stable supply voltage to the microcontroller, even as the battery voltage drops during discharge.

Why this component was chosen:

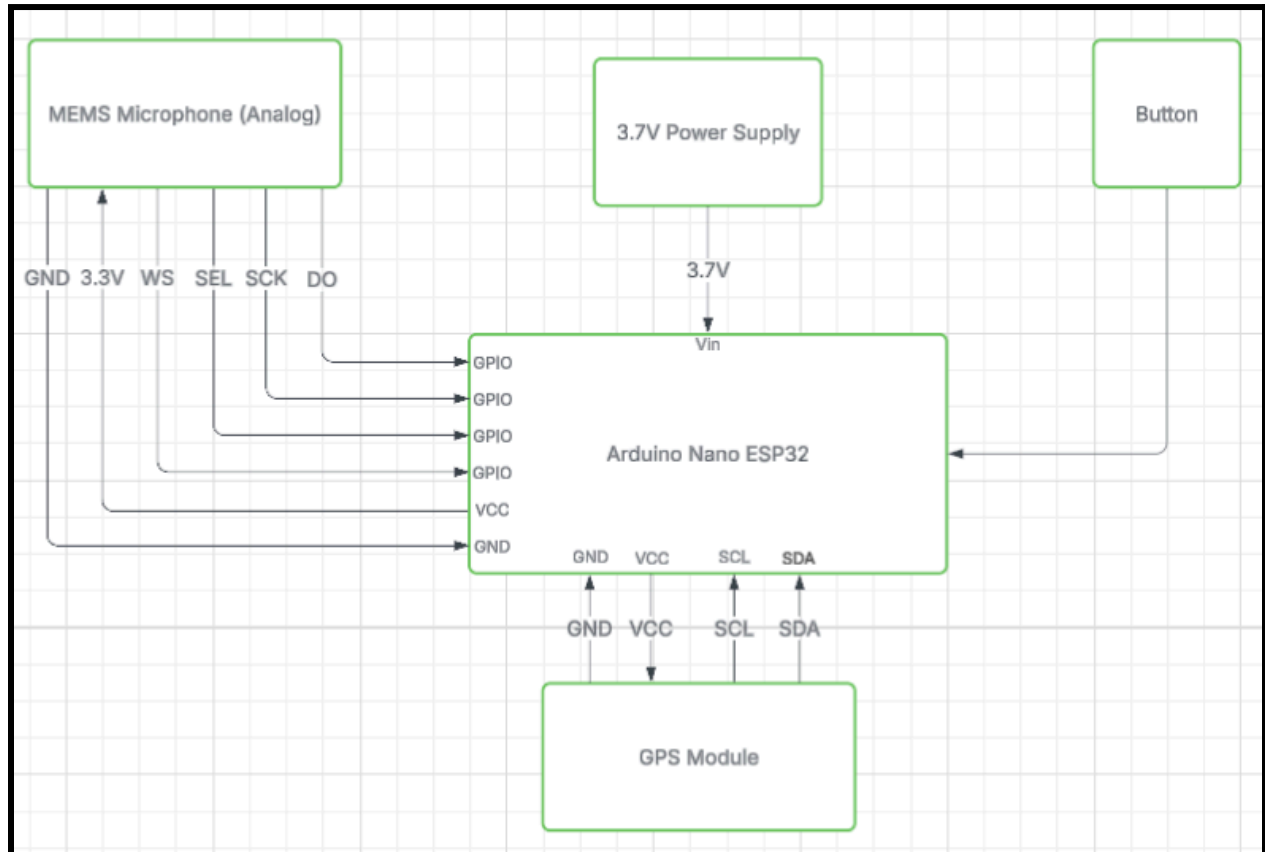
The MIC2288 was chosen as it's a compact and efficient boost converter that requires few external components, making it ideal for tight-spaced designs. It can provide a 5V output from a low-voltage battery while maintaining efficiency, maximizing battery life. Since the component is adjustable to 34V, this makes it easy to modify if the output voltage needs to be changed.

NOTE: The MIC2288 is not included within the final design of the device nor the schematic. Multiple voltages and external component values were tested, but the output voltage ripple was too much for the Arduino Nano ESP32 to handle such that the Bluetooth feature was not operational. Luckily, the entire device works perfectly when powered directly from the 3.7V battery into the microcontroller's VIN pin.

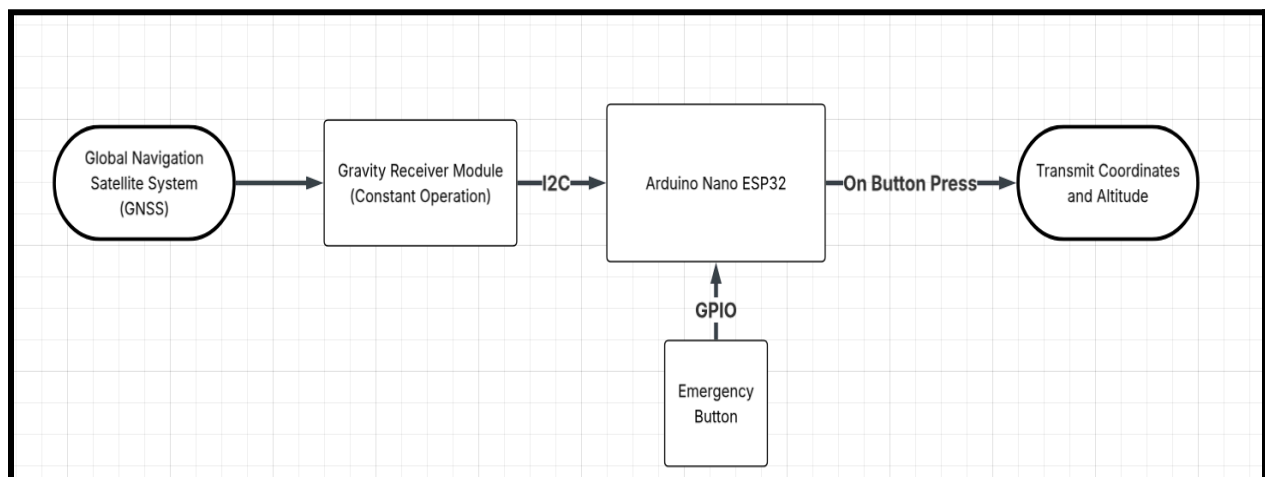
Hardware Documentation

Block Diagram

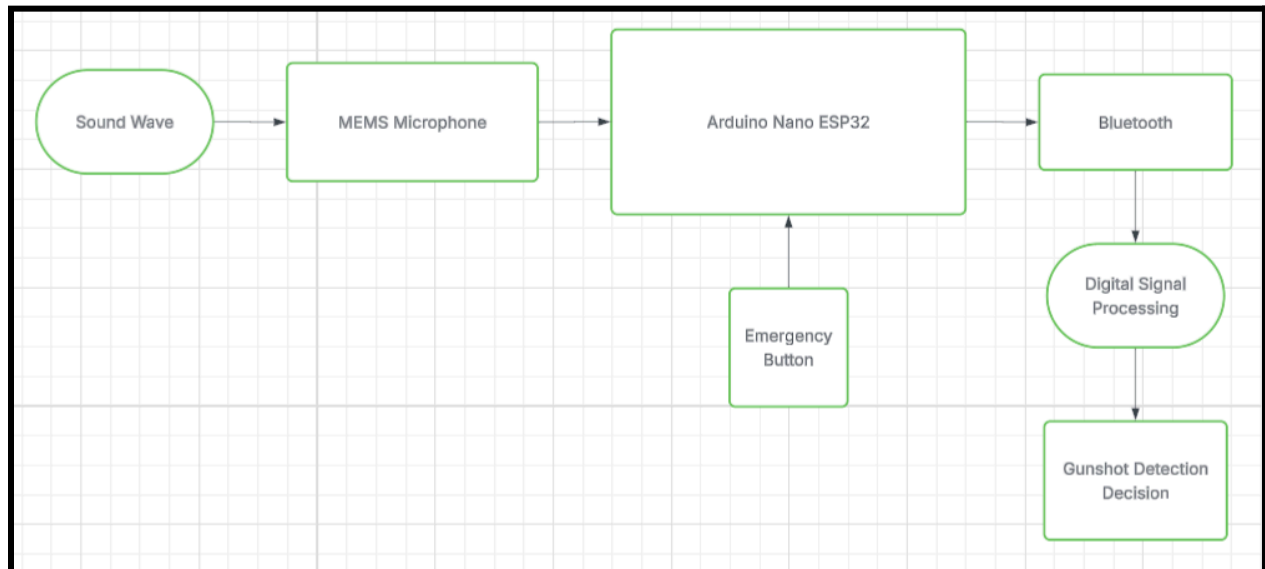
Overall System:



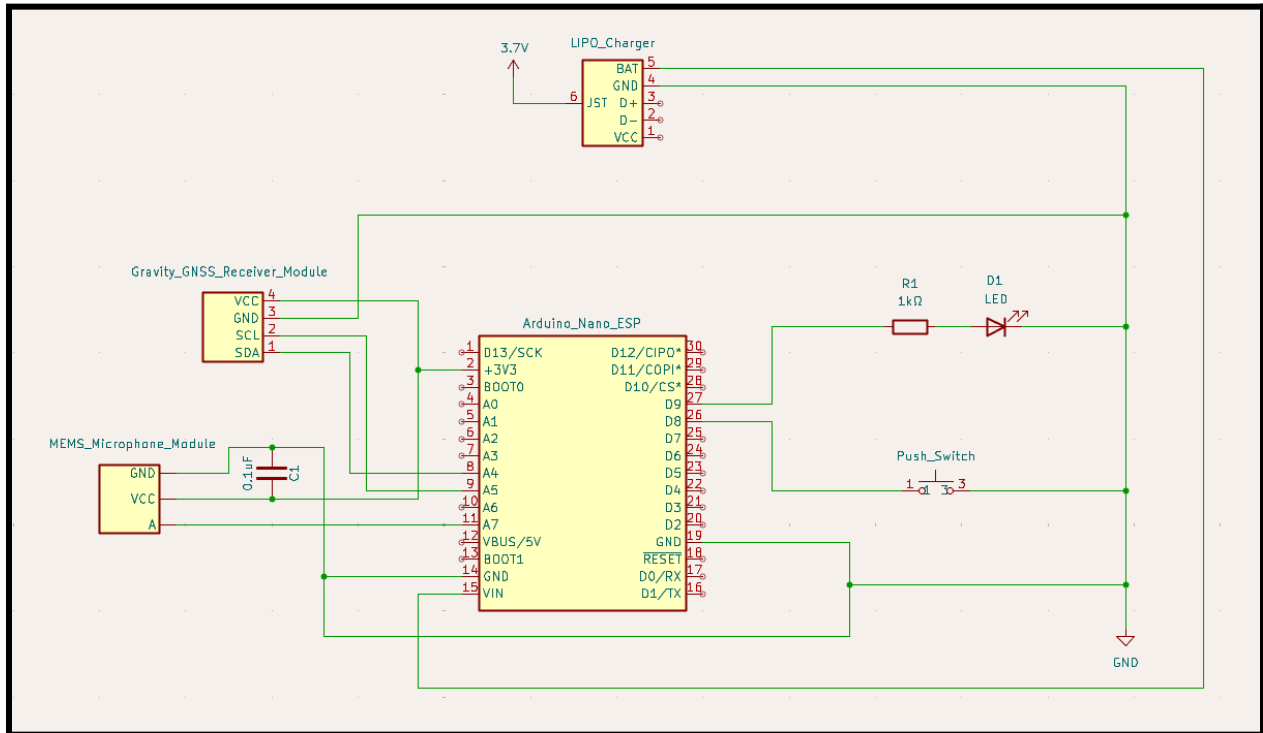
Position Module Visual:



Microphone Visual:



Schematics



Starting with the microphone, the VCC is connected to the 3.3V rail of the Arduino Nano ESP32. The analog pin of the microphone is connected to pin A7 of the ESP32. Between the GND and the VCC of the microphone, there is a decoupling capacitor of 0.1uF to limit the noise from the analog microphone. Next with the Gravity: GNSS Receiver Module (GPS Module), the VCC is connected to the same 3.3V rail of the ESP32, the SCL is connected to A5 of the ESP32 because the pin has SCL capabilities, and the SDA is connected to A4 of the ESP32 because the pin has SDA capabilities. The 3.7V is connected to the Micro Lipo Charger via a JST connector and then the Micro-Lipo Charger's BAT is connected to the Vin of the ESP32. The testing LED's anode is connected to a limiting resistor to pin D9 of the ESP32 and the cathode is connected to GND. The push switch pin 1 is connected to pin D8 of the ESP32 and pin 3 of the push button is connected to GND. All GND from all the components in this circuit connects to the common GND.